CSE 4165 FINAL EXAM CHEAT SHEET • SIDE 1: JAVASCRIPT (Q1) & PHP LOGIC (Q2)

Time: 1 hr 55 min | 30 Marks (Q1: 10M, Q2: 10M, Q3: 10M) | Strategy: Q1 30m • Q2 35m • Q3 45m • Check 5m

A4 DUPLEX • SIDE 1

Q1 SKELETON: HTML + JS FULL BOILERPLATE (10 MARKS) — JAVASCRIPT

```
<!— index.html —>
<input type="text" id="itemName" placeholder="Name">
<input type="number" id="itemVal" step="any" placeholder="Value">
<button id="actionBtn">Add / Calculate</button>
<label id="outLabel"></label>
<script src="script.js"></script>
```

```
// script.js
const GOAL = 2000; // Target from question (e.g. 2000, 20, 100)
let total = 0, count = 0;
const historyList = []; // Keeps track of entries/readings

const inputName = document.getElementById("itemName");
const inputVal = document.getElementById("itemVal");
const actionBtn = document.getElementById("actionBtn");
const outLabel = document.getElementById("outLabel");

function handleAction() {
  const name = inputName ? inputName.value.trim() : "";
  const val = parseFloat(inputVal.value);
  if (isNaN(val) || val ≤ 0) {
    alert("Please enter a valid positive number");
    return;
  }
  count++;
  total += val;
  if (name) historyList.push(name);
  const avg = total / count;

  // 4-Tier Ladder (Match exact wording in question)
  let feedback = "";
  if (total ≤ 800)      feedback = "You're off to a healthy start!";
  else if (total ≤ 1600) feedback = "Good progress, keep it balanced!";
  else if (total < GOAL)  feedback = "Almost at your limit!";
  else                    feedback = "Goal reached! Stay mindful!";

  // Exam Trap: Warning if user exceeds attempts/entries without reaching goal
  if (count > 10 && total < GOAL) {
    feedback += " Be cautious of frequent snacking!";
  }

  // Display inside label field
  outLabel.textContent = `Total: ${total} | Avg: ${parseFloat(avg.toFixed(2))} | Feedback: ${feedback}`;
  inputVal.value = "";
  if (inputName) inputName.value = "";
}
actionBtn.addEventListener("click", handleAction);
```

Q1 WORDING DECODER & FORMULA CHEAT BOX

Question Asks For...	JavaScript Exact Formula / Code
Running total / so far	`let total = 0; total += val;` (outside function)
Average so far	`const avg = total / count;`
Store entered items/names	`history.push(name); `[${history.join(", ")}]``
Number of entries / attempts	`let count = 0; count++;` on every valid click
Warning on > N entries without goal	`if (count > N && total < GOAL) msg += " Warning!";`
Format to 2 decimal places	`parseFloat(avg.toFixed(2))` (no trailing zeros: `+avg.toFixed(2)`)
Absolute difference |HR - 80|	`Math.abs(hr - 80)`
Password length pts (≥6 chars)	`p.length ≥ 6 ? Math.floor((p.length - 6) / 2) * 10 : 0`
Regex: Uppercase / Lower / Digit / Symbol	`/[A-Z]/.test(p)` • `/[a-z]/.test(p)` • `/[0-9]/.test(p)` • `/[!@#$%^&*]/.test(p)`
Average of an array	`const avg = arr ⇒ arr.reduce((a,b) ⇒ a+b, 0) / arr.length;`

Q2 SKELETON: HTML FORM + PHP LOGIC FULL BOILERPLATE (10 MARKS) — PHP

```
<!— form.html —>
<form action="calculate.php" method="POST">
  <label>Input 1: <input type="number" name="val1" step="any" required
></label><br>
  <label>Input 2: <input type="number" name="val2" step="any" required
></label><br>
  <label>Input 3: <input type="number" name="val3" step="any" required
></label><br>
  <input type="submit" value="Calculate">
</form>
```

```
<?php // calculate.php
// Always declare calculating function as specified by exam
function processLogic($v1, $v2, $v3) {
  // Pattern A: Ceiling Booking (venues, screens, battery modules)
  $unitsNeeded = ceil($v1 / $v3);            // CANNOT book partial: ceil()
  $totalCap    = $unitsNeeded * $v3;
  $emptyUnits  = $totalCap - $v1;            // Unused capacity / empty seats
  $wastedMoney = $emptyUnits * $v2;          // Wasted money based on rate

  // Pattern B: Target Comparison (items*days vs target or credits*courses)
  $totalProduced = $v1 * $v2;
  $targetDiff    = $totalProduced - $v3;
  if ($targetDiff == 0)      $targetMsg = "Target met exactly (0 difference)";
  elseif ($targetDiff > 0)  $targetMsg = "Above target by $targetDiff";
  else                       $targetMsg = "Below target by " . abs($targetDiff);

  // Pattern C: 4-Tier Performance / Efficiency Classification
  $status = "";
  if ($totalProduced ≥ 500)      $status = "Excellent";
  elseif ($totalProduced ≥ 300)  $status = "Good";
  elseif ($totalProduced ≥ 150)  $status = "Average";
  else                           $status = "Poor";

  // Efficiency % pattern: unused / totalCarried
  $unusedPct = ($totalCap > 0) ? ($emptyUnits / $totalCap) * 100 : 0;
  $effStatus = ($unusedPct ≤ 10) ? "Efficient" : (($unusedPct ≤ 25) ? "Acceptable" : "Wasteful");

  return [
    'units' ⇒ $unitsNeeded, 'empty' ⇒ $emptyUnits,
    'waste' ⇒ $wastedMoney, 'targetMsg' ⇒ $targetMsg, 'status' ⇒ $status
  ];
}

if ($_SERVER["REQUEST_METHOD"] == "POST") {
  $v1 = floatval($_POST['val1']);
  $v2 = floatval($_POST['val2']);
  $v3 = floatval($_POST['val3']);
  $res = processLogic($v1, $v2, $v3);

  // Render HTML Table matching exam sample output
  echo "<table border='1' cellpadding='6'>";
  echo "<tr><th>Input 1</th><th>Input 2</th><th>Input 3</th><th>Units</th><th>Empty/Unused</th><th>Result</th></tr>";
  echo "<tr><td>$v1</td><td>$v2</td><td>$v3</td><td>{$res['units']}</td><td>{$res['empty']}</td><td>{$res['status']}</td></tr>";
  echo "</table><br><a href='index.html'>Back</a>";
} ?>
```

Q2 WORDING DECODER & FORMULA CHEAT BOX

Question Keyword	Exact PHP Formula / Implementation
"Cannot book partial / minimum complete"	`$units = ceil($totalPeople / $capacity);`
"Empty seats / unused capacity"	`$empty = ($units * $capacity) - $totalPeople;`
"Money wasted on empty seats"	`$waste = $empty * $ticketPrice;` (from sample table)
"Target met exactly / above / below by"	`$d = $total - $target; if($d==0) … elseif($d>0) … else abs($d)`
"Best two CT marks out of three"	`$bestAvg = (($ct1 + $ct2 + $ct3) - min($ct1, $ct2, $ct3)) / 2.0;`
"Passed if total > 54 otherwise Failed"	`$status = ($total > 54) ? "Passed" : "Failed";`
"Unused energy ≤ 10% of total carried"	`$pct = ($unused / $totalCarried) * 100; if ($pct ≤ 10) …`
Number format with commas (e.g. 8,400)	`number_format($amount)` or `number_format($amount, 2)`

Exam Tip: Story distractor costs (e.g. "venue costs 15,000 BDT") vs sample output: check the table! In 251 & 252, Wasted Money = `empty_seats * cost_per_person`.

CSE 4165 FINAL EXAM CHEAT SHEET • SIDE 2: PHP & MYSQL (Q3) & QUERY TRANSLATION

Total Marks: 10M • Hand-in: database.sql (50%) + solution.php (50%) • Full PHP Code Required

A4 DUPLEX • SIDE 2

Q3 PART 1: database.sql SKELETON (TABLE + SAMPLE DATA)

SQL SETUP

```
CREATE DATABASE IF NOT EXISTS uiu_exam;
USE uiu_exam;

DROP TABLE IF EXISTS records;
CREATE TABLE records (
  id          INT PRIMARY KEY,
  name        VARCHAR(100),
  category    VARCHAR(50),
  price       DECIMAL(10, 2),
  stock       INT,
  rating      DECIMAL(3, 1),
  status      VARCHAR(50)
);

INSERT INTO records (id, name, category, price, stock, rating, status)
VALUES
(1, 'Item One', 'Electronics', 450.00, 10, 4.8, 'Active'),
(2, 'Item Two', 'Electronics', 250.00, 5, 4.2, 'Active'),
(3, 'Item Three', 'Furniture', 850.00, 8, 3.9, 'Pending'),
(4, 'Item Four', 'Furniture', 120.00, 15, 4.5, 'Overdue');
```

SQL Data Types: `INT` whole numbers • `VARCHAR(100)` text • `DECIMAL(10,2)` money/fee • `DECIMAL(3,1)` rating 4.5 • Double single-quotes for apostrophe: `'Cox''s Bazar'`

Q3 PART 2: solution.php FULL EXAM SKELETON

PHP & MYSQL

```
<?php
// 1. Database Connection
$conn = new mysqli("localhost", "root", "", "uiu_exam");
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

// 2. Reusable Helper: Executes SELECT and displays an HTML Table
function showQuery($conn, $title, $sql) {
  echo "<h4>$title</h4>";
  $result = $conn->query($sql);
  if (!$result) { echo "Error: " . $conn->error; return; }
  if ($result->num_rows > 0) {
    echo "<table border='1' cellpadding='5'><tr>";
    while ($f = $result->fetch_field()) echo "<th>{$f->name}</th>";
    echo "</tr>";
    while ($row = $result->fetch_assoc()) {
      echo "<tr>";
      foreach ($row as $val) echo "<td>$val</td>";
      echo "</tr>";
    }
    echo "</table>";
  } else { echo "0 results found.<br>"; }
}

// 3. Reusable Helper: Executes UPDATE / DELETE queries
function executeUpdate($conn, $title, $sql) {
  echo "<h4>$title</h4>";
  if ($conn->query($sql) === TRUE) {
    echo "Updated successfully. Rows affected: " . $conn->affected_row
s . "<br>";
  } else { echo "Update error: " . $conn->error . "<br>"; }
}

// Task 1: Category Aggregation
$sql1 = "SELECT category, COUNT(*) AS total_items FROM records GROUP B
Y category";
showQuery($conn, "Task 1: Category Count", $sql1);

// Task 2: Conditional Update with WHERE ... AND ...
$sql2 = "UPDATE records SET status = 'Grace Period' WHERE status = 'Ov
erdue' AND stock < 10";
executeUpdate($conn, "Task 2: Status Update", $sql2);

// Task 3: Bonus / Percentage Update with Upper Ceiling Cap
$sql3 = "UPDATE records SET price = price * 1.10 WHERE price > 400 AND
(price * 1.10) <= 1000";
executeUpdate($conn, "Task 3: Price +10% with Cap", $sql3);

// Task 4: Advanced SELECT with Sorting or Category Average Subquery
$sql4 = "SELECT name, category, price,
         CASE WHEN price > (SELECT AVG(price) FROM records r2 WHERE r
2.category = r1.category)
              THEN 'Top Seller' ELSE 'Regular Seller' END AS seller_st
atus
         FROM records r1 ORDER BY price DESC";
showQuery($conn, "Task 4: Category Average Comparison", $sql4);

// Single Value: Total Net Worth
$resNet = $conn->query("SELECT SUM(price * stock) AS net_worth FROM re
cords");
$rowNet = $resNet->fetch_assoc();
echo "<p><b>Net Worth: </b>" . number_format($rowNet['net_worth'], 2)
. " BDT</p>";

$conn->close();
?>
```

EXAM QUESTION WORDING → SQL COMMAND DICTIONARY

When Question Asks:	SQL Command to Write:
"Show / Display / Find / List"	`SELECT col1, col2 FROM table`
"For each category / department / region"	`GROUP BY CategoryName` (must appear in SELECT too)
"Total number of ... in each ..."	`SELECT col, COUNT(*) AS total FROM t GROUP BY col`
"Only include groups having more than N / avg > X"	`HAVING COUNT(*) > N` or `HAVING AVG(rating) > 4.4`
"Sorted largest first / highest to lowest"	`ORDER BY col_name DESC`
"Sorted smallest first / ascending"	`ORDER BY col_name ASC`
"Name contains word 'Beach' or 'Forest'"	`WHERE name LIKE '%Beach%'`
"Rating between 4.0 and 4.5 (inclusive)"	`WHERE rating BETWEEN 4.0 AND 4.5`
"If salary < 40,000 and rating != 'D', change to 'C'"	`UPDATE t SET rating='C' WHERE salary < 40000 AND rating <> 'D'`
"Add 5,000 bonus, only if resulting ≤ 60,000"	`UPDATE t SET sal = sal + 5000 WHERE sal > 50000 AND (sal + 5000) ≤ 60000`
"Add 10% bonus revenue / Increase by 10%"	`UPDATE t SET rev = rev * 1.10 WHERE rev > 70000`
"Reduce price by 10% / 10% discount"	`UPDATE t SET price = price * 0.90 WHERE category = 'Language'`
"Total estimated revenue: sum of fee × visitors"	`SELECT region, SUM(fee * visitors) AS rev FROM t GROUP BY region`
"Total net worth of entire bookshop"	`SELECT SUM(price * stock) AS total_worth FROM book_info`
"Label 'Top Seller' if above own category avg"	`CASE WHEN price > (SELECT AVG(price) FROM t t2 WHERE t2.cat = t1.cat) THEN 'Top Seller' ELSE 'Regular' END`
"Change two columns at once"	`UPDATE t SET status='Grace Period', fee=0.00 WHERE status='Overdue' AND days < 7`

CRITICAL EXAM RULES & ZERO-CRASH CHECKS

- **TRAP 1:** Never use `WHERE COUNT(*) > 1` or `WHERE AVG(x) > 4`! Aggregates MUST go in `HAVING` after `GROUP BY`.
- **TRAP 2:** Always run tasks in the **exact order listed**! Earlier `UPDATE` queries change rows for subsequent `SELECT` queries.
- **TRAP 3:** Double check column spelling! Case sensitivity in SQL depends on OS. Use exact names from the exam sheet.
- **TRAP 4:** In XAMPP, open PHP files via `http://localhost/web_final/ ...` (never double-click the file, or you get raw PHP text).
- **TRAP 5:** Blank white page in PHP? You missed a semicolon `;` or closing quote/bracket. Open `php.ini` display_errors or check syntax.
- **TRAP 6:** JS button does nothing? Press `F12` → Console. Ensure HTML input `id` matches `document.getElementById()`.
- **ZIP & SUBMIT:** 5 minutes before deadline, stop editing, zip `web_final`, and upload to eLMS!